Higher National Diploma in Engineering (Electrical)

First Year, Second Semester Examination — 2020

EE 1212-Introduction to Computer Systems

Question 01

I. Speaker, Printer and Multimedia projector are three devices that can be connected to a computer. a) Briefly explain the purpose of each device. (6 marks) b) Mention a port available in the computer relevant to each device. (3 marks)

II. Name the port shown in this picture. What is the purpose of having this port in a lap top computer? (3 marks)

III. What is Cache memory? Explain the difference in between L1 cache and L2 cache. (5 marks)

IV. Briefly explain the functionalities of CU and ALU. (8 marks)

Question 02

I. Briefly explain the term "**Computer Bus**". (4 marks)

II. Explain the functions of the following busses (6 marks) a) Data bus b) Address bus c) Control bus

III. Draw a block diagram showing the internal structure of a hard disk and name the components. (7 marks)

IV. Briefly explain **impact printers** and **non-impact printers**. Give two examples for non-impact printers. (5 marks)

V. Four main criteria are considered in evaluating printers. List three of them. (3 marks)

Question 03

I. Software can be acquired in different ways. Briefly explain **Freeware**, **Open source software** and **commercial software**. (6 marks)

II. **Compiler**, **loader** and **linker** are examples for system software. Mention the functionality for each of them. (6 marks)

III. Briefly describe the difference between '**Blog**' and '**Wiki**'. (4 marks)

IV. Briefly explain the use of **web browser** by giving two examples. (4 marks)

V. Mention one thing that you can create, using each of the following software tools (5marks) a) MS Excel b) MS Word c) MS Access d) MS PowerPoint e) Photoshop

Question 04

I. What is a hand-held device in computing? Give three examples for hand-held devices. (5 marks)
II. Mention three advantages of electronic mailing system compared to traditional mailing system. (3 marks)
III. Using a suitable diagram explain the client-server model in networking. (5 marks)
IV. Convert the following binary numbers into decimal equivalent (6 marks)
    a) 11100.10 b) 100011.0011 c) 11111.1100
V. Perform the following operation using two's complement notation (6 marks)
    a) -46+78 b) -26-17

Question 05

I. Expand the abbreviation **ADSL**. (2 marks)
II. ADSL is more suitable than other connection methods for places like ATIs. Give two reasons for this. (4 marks)
III. It is required to establish a network using the devices given below. Existing ADSL connection should be used for both voice calls and Internet facility. * Server computer - 01 * Desktop computers - 15 * Fire wall - 01 * Printer - 01 * Document Scanner - 01

The network needs to satisfy following requirements:

- All the desktop computers should have the Internet connection through the server.
- Printer and scanner should be accessed by all desktop computers.
    a) List the hardware resources and types of cables with connectors required to establish the above network. (5marks)
    b) What is the most suitable topology for this network? Explain the reason. (4 marks)
    c) Draw the network diagram including all the components you identified in part (a). (5 marks)
    d) What is the purpose of including a firewall in this network? (3marks)
    e) Mention an operating system to be installed in the server computer. (2 marks)